

BERLIN TEMPELHOF, Germany

WMO: 103840

Lat: 52.4675N

Lon: 13.4022E

Elev: 49

StdP: 100.74

Time Zone: 1.00 (EUC)

Period: 86-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.3	-8.3	-14.6	1.1	-10.8	-11.5	1.4	-7.0	12.4	7.2	11.1	6.3	3.3	80	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	30.2	19.1	28.2	18.4	26.4	17.7	20.5	26.9	19.7	25.6	18.7	24.3	3.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	22.4	17.7	12.8	21.9	16.8	12.0	21.1	59.1	27.4	56.3	25.6	53.3	24.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.7	7.8	DB	-11.7	34.1	5.3	2.2	-15.5	35.6	-18.6	36.9	-21.5	38.1	-25.4	39.7	(4)
(5)			WB	-12.2	22.4	5.1	1.0	-15.9	23.1	-18.8	23.7	-21.7	24.3	-25.4	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	1.4	1.3	4.9	9.3	14.2	17.4	19.7	18.6	15.1	10.4	5.2	2.5	(6)
(7)		DBStd	7.69	5.40	4.77	4.48	3.84	3.92	3.73	3.43	3.20	2.97	3.64	3.42	3.87	(7)
(8)		HDD10.0	1162	268	243	163	58	7	0	0	0	1	41	147	234	(8)
(9)		HDD18.3	3182	526	476	416	271	137	61	24	37	104	246	393	491	(9)
(10)		CDD10.0	1185	1	1	6	37	138	223	301	266	154	53	3	1	(10)
(11)		CDD18.3	163	0	0	0	0	9	34	67	44	8	0	0	0	(11)
(12)		CDH23.3	1333	0	0	0	5	92	278	576	319	63	0	0	0	(12)
(13)		CDH26.7	372	0	0	0	0	16	66	190	96	5	0	0	0	(13)
(14)	Wind	WSAvg	3.9	4.7	4.4	4.4	3.8	3.7	3.5	3.6	3.4	3.6	3.7	4.1	4.3	(14)
(15)	Precipitation	PrecAvg	576	46	34	39	36	53	66	63	60	45	41	47	49	(15)
(16)		PrecMax	835	119	90	117	88	147	193	185	160	135	144	102	130	(16)
(17)		PrecMin	376	6	2	9	1	4	2	7	7	5	4	1	10	(17)
(18)		PrecStd	108	24	21	22	21	31	42	35	33	26	29	21	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.4	18.0	24.0	29.1	31.0	33.5	32.9	27.2	21.6	14.1	12.9	(19)
(20)			MCWB	9.9	10.5	11.6	13.6	18.6	18.7	19.6	20.7	18.1	14.9	11.5	10.6	(20)
(21)		2%	DB	10.8	11.4	15.2	20.9	25.6	28.6	30.9	29.2	25.2	19.0	12.2	11.0	(21)
(22)			MCWB	8.6	8.5	9.8	12.9	16.2	18.1	19.2	19.1	17.0	14.2	10.3	9.0	(22)
(23)		5%	DB	9.0	9.0	13.0	18.2	23.8	26.5	28.8	26.8	23.0	17.5	11.1	9.8	(23)
(24)			MCWB	7.4	6.8	9.0	11.4	15.1	17.5	18.7	18.2	16.4	13.3	9.4	8.1	(24)
(25)		10%	DB	7.8	7.2	11.2	16.0	21.9	24.5	26.8	24.5	20.7	15.9	10.0	8.2	(25)
(26)			MCWB	6.1	5.4	7.8	10.4	14.2	16.8	18.1	17.6	15.6	12.4	8.6	6.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	10.9	12.4	14.4	19.6	21.0	21.9	21.4	19.6	16.0	12.0	10.8	(27)
(28)			MCD B	12.3	14.0	16.5	21.1	25.5	27.0	28.4	29.8	25.5	19.5	13.7	12.8	(28)
(29)		2%	WB	8.9	8.7	10.8	13.3	17.9	19.7	20.7	20.2	18.2	14.8	10.8	9.2	(29)
(30)			MCD B	10.4	11.1	14.1	19.4	23.0	25.3	27.0	26.3	22.2	18.5	12.1	10.8	(30)
(31)		5%	WB	7.4	7.1	9.6	12.1	16.3	18.5	19.8	19.3	17.1	13.7	9.8	8.0	(31)
(32)			MCD B	8.7	8.7	12.4	17.7	21.8	24.1	26.2	24.7	21.0	16.6	11.0	9.5	(32)
(33)		10%	WB	6.1	5.5	8.2	10.8	14.9	17.5	18.7	18.4	16.0	12.9	8.8	6.8	(33)
(34)			MCD B	7.4	6.9	10.9	15.1	20.7	23.2	25.0	23.3	20.5	15.5	9.8	8.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	6.8	8.6	9.4	9.0	9.1	8.5	8.1	7.3	4.8	4.1	(35)
(36)			MCDBR	5.0	7.7	10.1	12.5	12.6	12.1	12.8	12.6	12.3	9.6	5.4	5.1	(36)
(37)			MCWBR	4.4	5.5	6.0	6.4	5.7	4.9	4.8	5.3	5.6	5.6	4.2	4.2	(37)
(38)		5% WB	MCD BR	5.0	7.4	9.3	10.9	11.2	10.8	10.8	10.2	9.6	8.3	5.1	5.1	(38)
(39)			MCWBR	4.5	5.6	6.0	6.0	5.7	5.0	4.9	4.8	5.3	5.3	4.3	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.353	0.394	0.419	0.416	0.413	0.434	0.427	0.389	0.376	0.351	0.321	(40)	
(41)		taud	2.310	2.285	2.193	2.197	2.250	2.282	2.236	2.268	2.350	2.346	2.351	2.323	(41)	
(42)		Ebn at Noon	651	744	782	812	836	841	813	796	785	709	617	589	(42)	
(43)		Edh at Noon	64	86	115	129	129	126	130	119	98	81	62	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.63	1.33	2.35	3.93	4.87	5.18	4.85	4.17	3.04	1.69	0.77	0.47	(44)	
(45)		RadStd	0.07	0.23	0.26	0.51	0.58	0.41	0.64	0.33	0.35	0.20	0.10	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page